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Control of combustion instability with a high-momentum air-jet

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Abstract

An improved strategy for controlling combustion oscillations using a high-momentum air-jet that changes the reactant mixing process is presented in this paper. The oscillations in the swirl-stabilized spray combustor of interest are dominated by an acoustic mode (235 Hz) with a low-frequency (13 Hz) bulk mode (of the upstream cavity) oscillation superimposed. The bulk mode gives rise to an axial movement of the flame and leads to cyclic variations of the dominant mode time period. These cyclic variations lead to phase jitter and make control more difficult. The most effective strategy for maintaining control is shown to be achieved through the use of a new concept which utilizes a high-momentum air-jet injected directly into the regions of positive Rayleigh Index from where the instability originates. Improvements with this new strategy relative to the traditional fuel-modulation-based control are demonstrated.
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Keywords: Combustion oscillations; High-momentum air-jet; Active control for low-frequency oscillation (13 Hz)

1. Introduction

Large-amplitude combustion oscillations, driven by a coupling of the longitudinal acoustic mode of the combustion system with the unsteady heat release fluctuations, leads to performance degradation, unacceptable noise levels, and structural damage [1–3]. All ducted combustion devices including propulsion systems, aero-engines, and industrial gas turbines are susceptible to such combustion oscillations. Therefore, feedback control has been suggested as a means for providing robust operation of combustors operating over a wide range of equivalence ratio. In this paper, attention is focused on control of combustion oscillations in a nonpremixed swirl-stabilized spray combustor, with the intent of finding strategies that provide the most effective control. To support this goal, detailed heat release measurements (chemiluminescence and CH imaging), Mie scattering imaging from the fuel spray, and pressure measurements have been made to identify the basic mechanisms and to utilize this information for effective control.

Control of longitudinal or bulk mode combustion oscillations have been studied by many investigators, but in these studies attention has been primarily focused on controlling the dominant instability mode. However, in many situations, the combustion oscillations are characterized by a dominant acoustic mode modulated by a low-frequency mode of less than 50 Hz. The presence of a low-frequency mode can

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give rise to an axial movement of the flame, which, in turn, can lead to spatiotemporal modulations in the heat release dynamics driving the instability. Such situations exist in practical combustors and have been reported [4–10] but have received little attention especially in relation to combustion instabilities. Presence of two modes (e.g., an acoustic mode around 300 Hz and a bulk mode of less than 50 Hz) have been reported by Cohen et al. [4] in a lean premixed single-nozzle combustor and at realistic engine operating conditions, Poinsot et al. [5] in a nonpremixed multiple-flame combustor, Isella et al. [6] in a premixed dump combustor, Langhorne [7] in ethylene/air premixed flame stabilized in the wake of a conical gutter, and De Zilwa et al. [8] in ducted round dump combustors with different sizes of the upstream cavities. Both Bradley et al. [9] and Gutmark et al. [10] reported flame movement, and even detachment and reattachment, in a swirl burner due to the presence of low-frequency oscillations. In all these studies, the primary emphasis has been on active control directed at suppressing the dominant mode oscillations, and little attention has been paid to the low-frequency mode.

Recently the mechanism of combustion oscillations associated with the operation of lean-premixed combustion close to the extinction limit was investigated by De Zilwa et al. [11,12] and Uhm [13]. These studies described in detail the flame movement at a frequency much lower than the acoustic mode frequency. The oscillations close to the extinction limits were observed to be related to the local extinction of the flame due to high strain rates close to the expansion plane followed by relight at downstream locations where the strain rate was smaller. These oscillations were greatly amplified when the duct exit was constricted with a nozzle, due to the coupling between the bulk mode and the low-frequency oscillations. Combustion oscillations, close to stoichiometry, with acoustic or bulk modes modulated by low-frequency were also investigated by Uhm [13], but efforts directed at active control were not successful. The inability to effectively control the instability is partly related to the coupling of the low-frequency and the dominant mode. This issue is specifically examined in this paper, and strategies to control oscillations under conditions where the acoustic mode is modulated by a low-frequency mode over a range of equivalence ratio are explored. Under such conditions, as shown by Uhm [13], traditional feedback control strategies are not as effective, and improved concepts for feedback control need to be explored.

The literature on feedback control of combustion instabilities is extensive and documented in the reviews by Culick [14], Schadow and Gutmark [15], Candel [16], and McManus et al. [17]. Active control has traditionally been implemented by acoustic drivers [18,19] and by pulsed fuel injection [20,21]. Sivasegaram and Whitelaw [22] and Sivasegaram et al. [23] showed that the suppression of the dominant frequency in flows with competing frequency modes led to the excitation of a rival frequency. Yu [24], Fleifil et al. [25], Wilson et al. [26], and Bhidayasiri et al. [27] found that control can be ineffective at high flow rates, due to bimodal combustion instabilities and a shift in the position of the flame.

The present experiments were performed in a swirl-stabilized spray combustor where a dominant acoustic mode was accompanied by a low-frequency oscillation that corresponds to the bulk mode of the upstream air-injection cavity. Under certain flow conditions, active control of the dominant combustion instability is shown to be relatively ineffective in this combustor with fuel modulations using an automotive fuel injector or a proportional-drive injector. In this paper, a new strategy using a modulated high-momentum air-jet (with a proportional-drive valve) is demonstrated and shown to be considerably more effective than the traditional fuel-modulation strategies. Detailed measurements to identify the heat release and droplet distribution patterns and to parametrically explore the effectiveness of the high-momentum air-jet were performed.

2. Experimental setup

The experiments were carried out in a multi-fuel-feed combustor operating at nearly 150 kW heat release. The combustor configuration, shown in Fig. 1, consists of two concentric large-area-ratio nozzles. Swirl vanes with 45° angles are placed in the inner and outer air streams at the exit of each nozzle. Each nozzle can be acoustically forced by an array of eight 75-W loudspeakers mounted at equal polar angles about the circumference of the nozzles. Eight liquid fuel atomizers (Arizona Mist; 0.3-mm orifices), equally spaced circumferentially and located between the coaxial jet streams, are provided. A fuel nozzle coupled to an automotive fuel injector is located at the geometric center of the inner air stream and used for fuel-modulation control. For the high-momentum control air-jet, the central fuel nozzle is replaced by a nozzle with straight-through holes of different sizes (1.5, 3, and 6 mm). For modulating the air stream, the air delivery tube is connected to a proportional-drive valve located outside the combustor (Fig. 1). The length of the tube between the nozzle and the proportional-drive valve is about 1 m. The valve responds to the input signal proportionally with a flat response up to a maximum frequency of 300 Hz. Com-

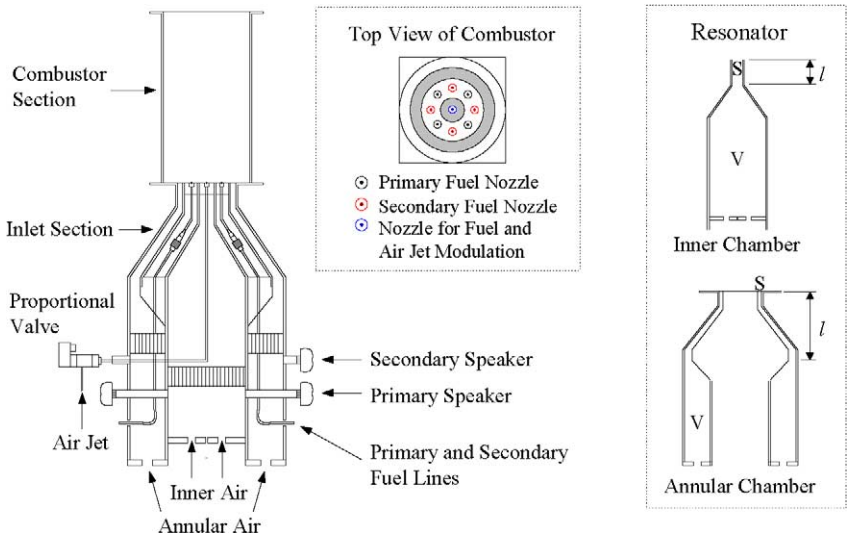


Fig. 1. Coaxial jet spray combustor.

pressed air is introduced upstream of the valve and adjusted with an air flow meter. Independent fuel supply lines are provided to the central injector and the two pairs of four unmodulated circumferential fuel nozzles. Ethanol is used as the liquid fuel and pressurized to a maximum of 18 bar in three fuel tanks by high-pressure nitrogen.

The combustor is equipped with a square shell with 19.5-cm sides and windows that are either stainless steel or quartz for optical access. A high-sensitivity water-cooled pressure transducer (Kistler 6061B) is mounted at an axial distance of 6 cm from the expansion plane to measure pressure oscillation and to provide the feedback signal for control. Light emission is recorded at the CH radical wavelength using a photodiode with a 430-nm centered-optical filter, and these measurements are assumed to be representative of the heat release fluctuations from the flame. The pressure and CH signals are processed in real time using a digital signal processor (DS1103; DSPACE; 333-MHz Motorola power PC) used in active control. The spatial and temporal variations in the CH chemiluminescence (heat release) are also visualized using a Princeton Instruments PI-MAX 512×512 ICCD camera with a UV lens (Electrophysics) and a bandpass filter (DIOP) at 430 nm. The images are triggered with respect to the pressure oscillations in the combustor to determine the phase-averaged distributions of the heat release at different instances of the pressure oscillation cycle. The distribution of fuel droplets is visualized using a SharpVISION 1300-DE 1280×1024 charge-coupled device (CCD) camera and a pulsed laser sheet light (New Wave Nd:Yag laser) synchronized together by a laser timing interface from Integrated Design Technologies.

3. Results and discussion

3.1. Characteristics of combustion oscillations

Fig. 2 shows the pressure signal at two locations (in the combustor and 10 cm upstream of the dump plane) and the CH light signal for flow conditions ($1.76\ \mathrm{m}^3/\mathrm{min}$ of air and $3.4\ \mathrm{g/s}$ of fuel; overall equivalence ratio of 0.89) leading to combustion instability. The spectra in Fig. 2b show a dominant frequency of 230 Hz modulated by a low-frequency of 13 Hz. The low-frequency modulation is due to the bulk mode of the air delivery chamber, and this mode and its harmonics are clearly evident in the pressure spectra recorded in the delivery chamber and the CH spectra recorded in the combustor. It can also be seen in the time traces of the pressure and CH signals (Fig. 2a) and plays a key role in the effectiveness of the control strategy. The bulk mode frequency of the upstream air delivery chamber is calculated by assuming it to be the resonant frequency of a Helmholtz resonator ($f = 1/2\pi\sqrt{c^2 S/Vl}$, where f, c, S, V, and l are the resonant frequency, sound speed, cross-sectional area, volume, and length of the resonator). For the annular air delivery chamber, the resonant frequency is calculated to be about 16 Hz. Given the uncertainties in the estimates of the volume and the exact air temperature in the delivery chamber, the estimated bulk mode frequency is considered to be close to the measured bulk mode frequency of 13 Hz. For the inner chamber the

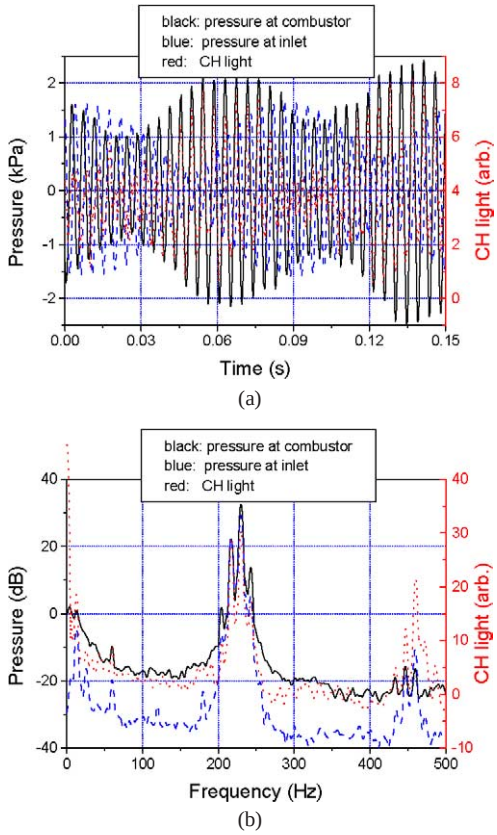


Fig. 2. Pressure signals in the combustor (black) and delivery chamber (blue-dash) and CH light emissions (red dot) under conditions of instability. Inner and annular airflow rates, 0.51 and 1.25 m^3/min, respectively. Primary and secondary fuel flow rate, 3.4 g/s. Overall equivalence ratio, 0.89.

bulk mode frequency is 28 Hz. The dominant mode also has associated side bands (Fig. 2b), and as will be seen later, the amplitude of these side bands are linked to the flow conditions in the combustor.

Pressure measurements at different locations in the air delivery chamber (Fig. 3) indicate a half-wave mode, while in the combustor (exit open to the atmosphere) a quarter-wave mode is obtained (not shown). Thus, a three-quarter-wave mode in the entire system is established, but the dominant mode is strongly influenced by bulk mode in the air delivery chamber, showing a clearly discrete peak (13 Hz) compared to that in the combustor in Fig. 2b. This is clearly seen in Fig. 2 and can be deduced from the amplitude of the rms of the total pressure fluctuation and the rms of the locus of the pressure peaks (or the rms of the low-frequency component of the pressure signal) at the pressure antinode near the dump plane, which are 0.8 kPa (Figs. 3 and 19a) and 0.35 kPa (Fig. 19b), respectively. This difference implies the existence of a second mode in the system.

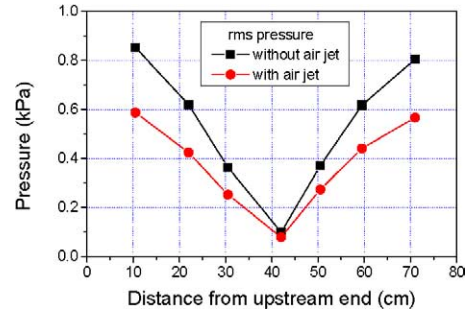


Fig. 3. Waveform of the dominant acoustic mode (230 Hz) and bulk mode (13 Hz) in the annular air stream delivery chamber. Same flow conditions as those in Fig. 2. Dump plane located at approximately 100 cm.

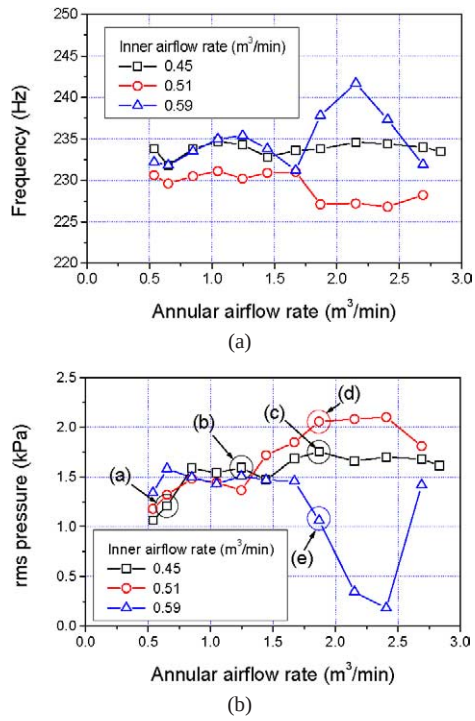


Fig. 4. Dominant frequency of combustion oscillation and its amplitude in flow conditions. (a)–(c) (marked on part b) 0.65, 1.25, and 1.87 m^3/min annular airflow rates, respectively. (d) and (e) (marked on part b) 0.51 and 0.59 m^3/min inner flow rates, respectively. Primary and secondary fuel flow rate, 3.4 g/s.

Fig. 4 shows the frequency and the amplitude of the large-amplitude combustion oscillations (greater than 1 kPa) as a function of the annular airflow rate for three different values of the inner-stream airflow rate. When the instability is greater than 1 kPa, the dominant frequency ranges from 227 to 235 Hz and is almost invariant with the flow rate or equivalence ratio. For unstable conditions, the pressure os-

cillation amplitude increases with the annular airflow rate (Fig. 4b). Note that for an inner airflow rate of 0.59 m^3/min, stable conditions are achieved for annular airflow rates in the range of 1.9–2.6 m^3/min, beyond which the instability levels rapidly rise.

Figs. 5a–5c show the pressure spectra in the combustor, the pressure spectra in the air delivery chamber, and the CH spectra in the combustor, respectively, for three different values of the annular airflow rates or overall equivalence ratios (in the range of 1 to 0.79). For clarity, the spectra in the lower-frequency range (0–50 Hz) are shown separately. In all cases, the spectra show a dominant mode around 230 Hz with significant side band modes of 230 ± 13 Hz in either direction and a low-frequency mode around 13 Hz. The low-frequency 13-Hz mode is clearly evident in the delivery chamber pressure spectra in Fig. 5b and in the CH spectra in Fig. 5c. The latter may be indicative of low-frequency flame movement as observed in De Zilwa et al. [11,12] and Uhm [13]. The side band mode is believed to be an outcome of the interaction between the dominant acoustic mode in the combustor and the bulk mode in the air delivery chamber. As the annular airflow rate is increased to 1.44 m^3/min, it alters the acoustic boundary condition at the exit of the delivery chamber and enhances

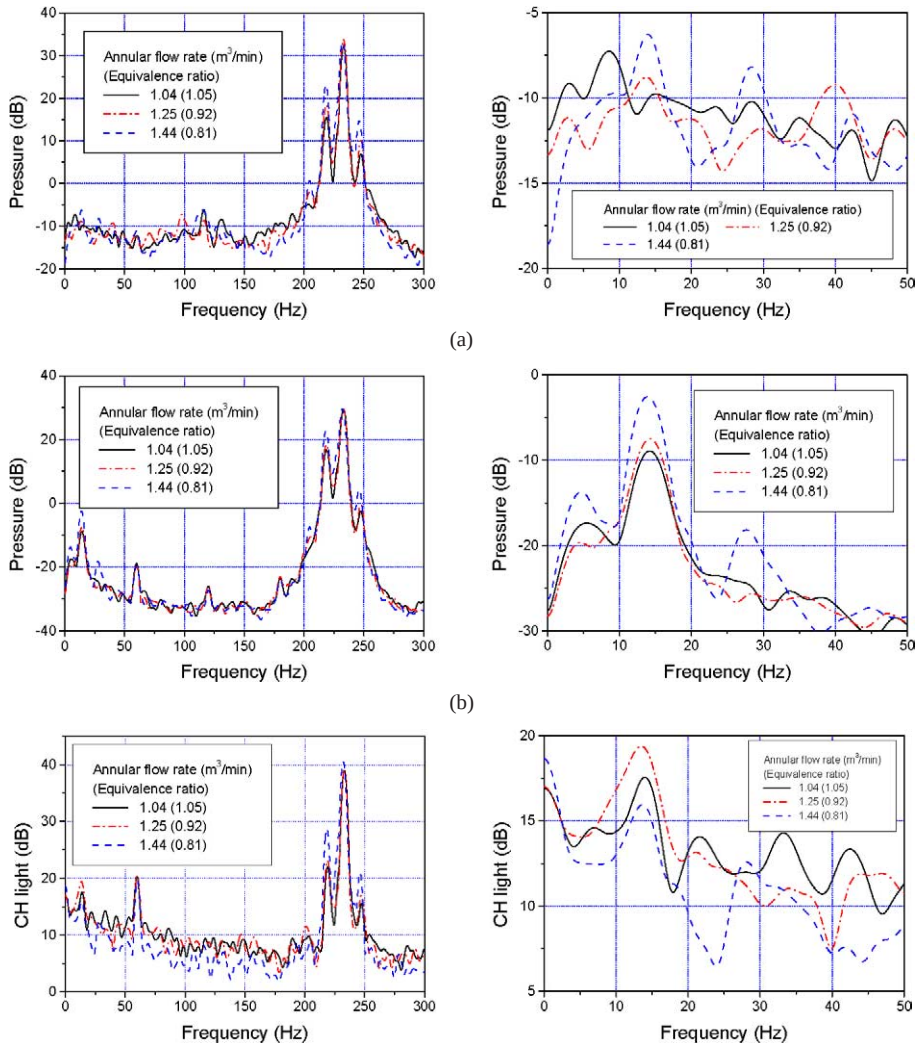
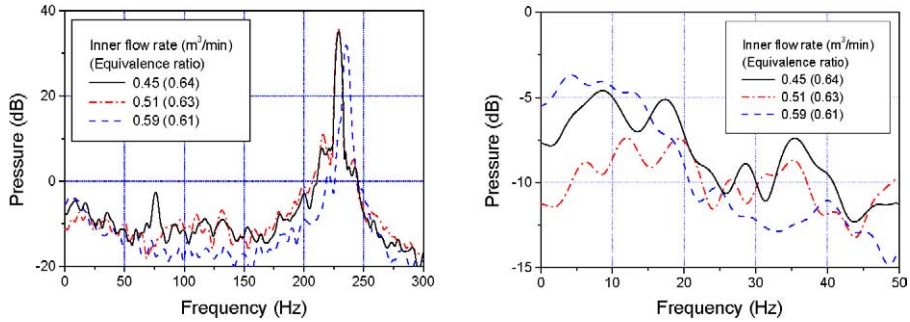
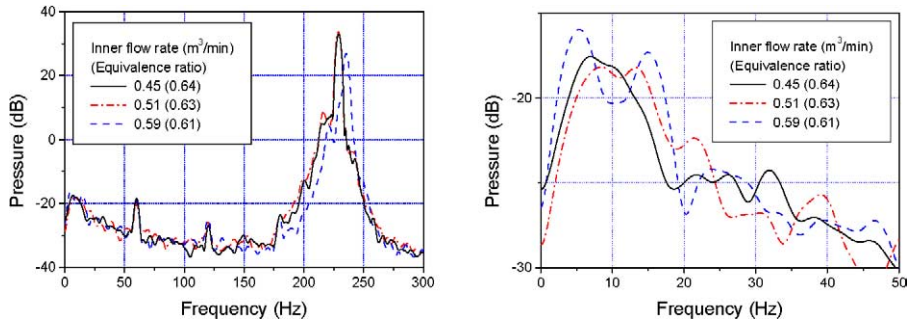


Fig. 5. Pressure and CH light spectra. Primary and secondary fuel flow rate, 3.4 g/s. (a)–(c) Pressure at the combustor and annular air delivery chamber and CH light. Inner airflow rate of 0.45 m^3/min and different annular flow rates of 1.04, 1.25, and 1.44 m^3/min (equivalence: 1.05, 0.92, and 0.81). (d) and (e) Pressure at the combustor and annular air delivery chamber, respectively. Annular airflow rate of 1.87 m^3/min and different inner flow rates of 0.45, 0.51, and 0.59 m^3/min (equivalence: 0.64, 0.63, and 0.61).



(d)



(e)

Fig. 5. (Continued.)

the bulk mode (Fig. 5b), leading to an increase in the peak in the amplitude of the side band instability frequency. This can also be seen in the time traces of the pressure signal at two different annular airflow rates (Figs. 6a and 6b). At lower annular airflow rates (1.25 m^3/min; condition (b) in Fig. 4b), the low-frequency 13-Hz modulation is clearly evident (Fig. 6a), and with higher annular airflow rate of 1.44 m^3/min (with the same inner airflow rate) the low-frequency 13-Hz modulation is even larger (Fig. 6b). However, as the annular airflow rate is increased beyond 1.44 to 1.87 m^3/min and the overall equivalence ratios move toward the lean limit, there are increases in the overall pressure rms for inner airflow rates of 0.45 and 0.51 m^3/min (see Figs. 4b and 6c) and a decrease in the amplitude of the low-frequency mode (seen clearly by comparing Fig. 5e with Fig. 5b). This reduction in the low-frequency mode is also accompanied by a reduction in the side bands (Fig. 5d). Further increase of the inner airflow rate to 0.59 m^3/min produces a decrease in the pressure oscillation levels as seen in Figs. 4b and 6d.

3.2. CH imaging and Mie scattering from droplets

Fig. 7 shows the time-averaged CH intensity distributions for increasing airflow rate conditions marked as (a)–(e) in Fig. 4b. As the annular airflow rate is increased (Figs. 7a–7c), the CH intensity levels are increased, leading to higher instability levels (Fig. 4b). Increasing the inner airflow rate (Figs. 7c–7e) results in an increase of the heat release intensity in Fig. 7d (and an increase in the pressure oscillation amplitude in Fig. 4b), but increasing the inner airflow rate further in Fig. 7e leads to a smaller reaction or a high-CH emission zone. The latter effect is likely to be linked to the local extinction due to straining of the flame or larger droplet dispersion at higher inner flow rates and leads to the reduction of the amplitude of the pressure fluctuations (Fig. 4b). Thus, local suppression of the flame (due to larger droplet spreading or higher straining) can lead to a reduction of the oscillations, and this observation can be utilized as a concept for achieving combustion control.

To better understand the dispersion characteristics of the fuel spray and how these are influenced by the inner and annular airflow rates, Mie-scattering images from a Nd:Yag pulsed-laser sheet (pulses separated by 10 ns) were obtained for different flow conditions and are shown in Figs. 8a–8f. These figures show the spray images on only one side of the circumferentially distributed spray injection pattern. These images result from the accumulation of 10 images during a 2-s window, with each image acquired with 0.2 s CCD camera exposure time. Fig. 8a shows the fuel spray without any air supply and shows roughly a 45° cone angle. Fig. 8b shows the fuel spray with inner

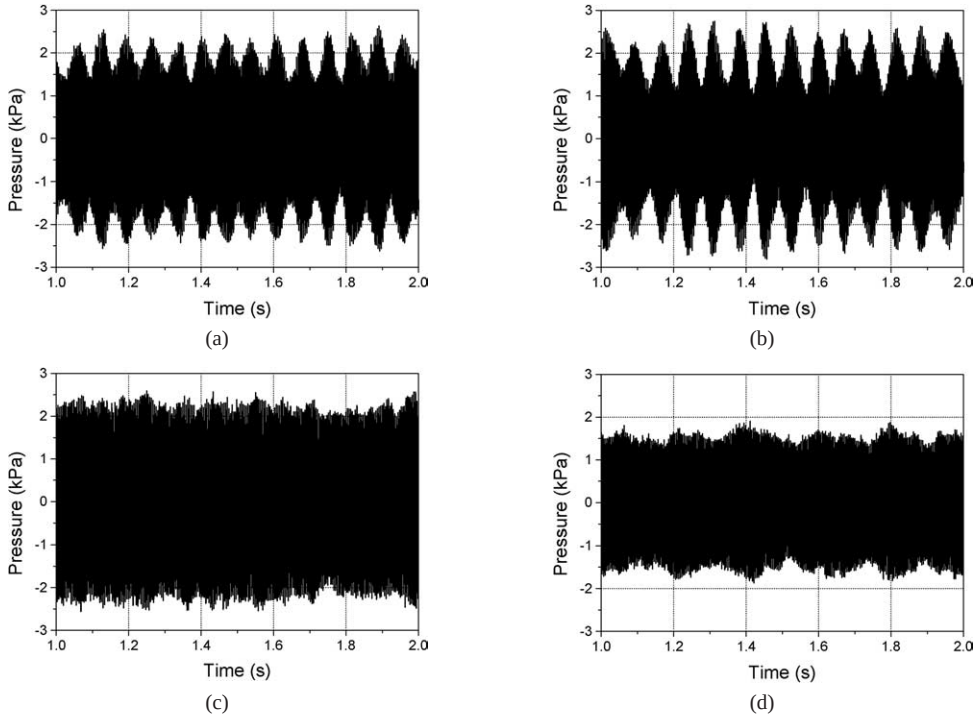


Fig. 6. Pressure signals at combustor. Primary and secondary fuel flow rate, 3.4 g/s. (a) and (b) Annular airflow rates of 1.25 and 1.44 m^3/min (equivalence: 0.89 and 0.79), respectively, and same inner airflow rate of 0.45 m^3/min. (c) and (d) Inner airflow rates of 0.45 and 0.59 m^3/min (equivalence: 0.64 and 0.61), respectively, and same annular airflow rate of 1.87 m^3/min.

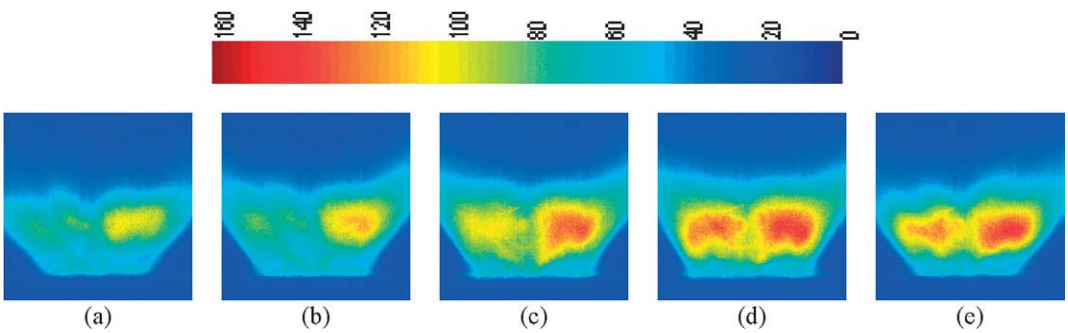


Fig. 7. Time-averaged distribution of CH radical intensity. (a)–(e) Same flow conditions as those shown in Fig. 4b where different flow conditions are marked by (a)–(e), respectively. Exposure time, 5 ms.

and annular supply of 0.45 and 0.65 m^3/min (corresponding to the condition marked (a) in Fig. 4b with low-amplitude pressure oscillations). For improved clarity and resolution in the images, the gray scale range 0 to 255 was reduced to 50 to 80 in Figs. 8b–8f, with all values below 50 set to 0, and all values above 80 set to 80. The dark region in Figs. 8b–8f indicates that the gray level is larger than 80 in this region and is due to the liquid phase close to the nozzle tip, while the bright region around its periphery reflects a high density of fuel droplets, indicating liquid fuel atomization and droplet breakup. Note that the inner and outer air flow rates are different in all cases, have swirl, and therefore generate droplets that are mostly inclined away from the centerline (right-hand boundary in each figure). In Fig. 8b, the fuel droplets are generally directed vertically upward, but a significant concentration of the droplets is also observed in the vicinity of the centerline. This case corresponds to the lowest inner and annular flow rates and has the largest spatial distribution of droplets but the weakest heat release (Fig. 7a) and pressure oscillation (Fig. 4b). The CH distributions in Fig. 7 indicate that the regions with high heat release (and

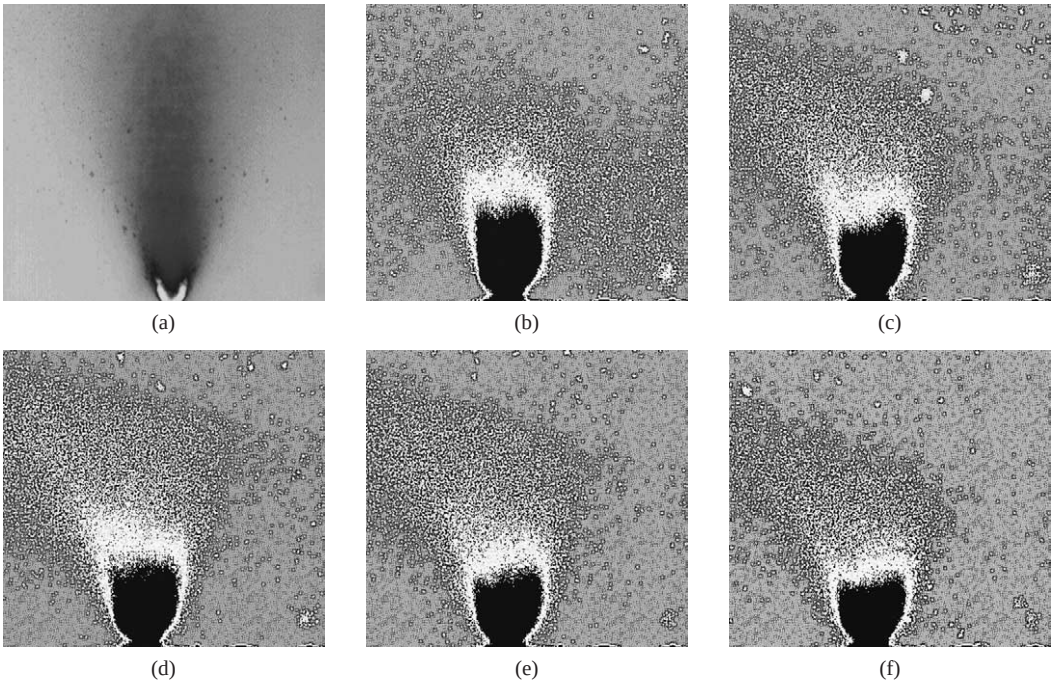


Fig. 8. Fuel spray visualization. (a) Without air. (b)–(f) Same flow conditions as those shown in Figs. 7a–7e, respectively.

combustion dynamics) are radially displaced from the centerline, and the centerline region itself has lower levels of heat release. Therefore, the droplet distribution pattern shown in Fig. 8b, where a significant concentration of the droplets is around the centerline, has lower heat release levels in the regions where the instability originates. With increasing annular airflow rates (Figs. 8c and 8d), the atomization region (the bright region directly above the upper end of the dark region where droplet breakup is occurring) appears to be closer to the dump plane, and the droplets are pushed radially outward to a greater extent. This implies greater concentration of droplets in the regions radially displaced from the centerline, and this correlates well with the CH images in Fig. 7 which show higher levels of CH distributions and greater radial spreading at the higher annular airflow rates. This is associated with increased amplitude of the pressure oscillations. With the annular flow rate held constant, increasing the inner flow rate (Figs. 8e and 8f) appears to decrease the presence of the droplets closer to the center and to decrease the vertical spreading of the droplets. This observation is consistent with the CH images in Figs. 7d and 7e where a reduction in the intensity and spatial extent (in the axial direction) of CH emission is observed at the highest inner airflow rates. As noted earlier, the reduction in CH levels and spatial extent leads to a corresponding reduction in the pressure oscillation amplitude (Fig. 4b).

Phase-locked images of CH are shown in Fig. 9 at phase angles of 10 and 175° of the pressure cycle for time periods with high-amplitude pressure oscillations (Figs. 9a and 9b) and for time periods with lower-amplitude pressure oscillations (Figs. 9c and 9d). High- and low-amplitude oscillations of the pressure are due to the low-frequency modulation, as clearly evident in the pressure time traces of Fig. 2. Thus, phase-averaging of the CH images can be performed separately for time periods with high amplitudes (Figs. 9a and 9b) and time periods with lower oscillation amplitudes (Figs. 9c and 9d). It is clearly evident that there is a strong correlation between the pressure and the CH variations, and the peaks in the pressure amplitudes are associated with the peaks in the CH levels. Furthermore, the variation in the CH amplitude is greater during the high-amplitude pressure oscillation cycles. The peak CH levels occur close to the dump plane, which corresponds to a pressure antinode.

Phase-locked images of CH are shown in Fig. 9 at two phase angles of 10° (close to the occurrence of the peak pressure) and 175° (close to the minimum pressure instance) during two instability time periods of the combustor operation. The first time interval corresponds to time instances in the vicinity of the low-frequency peak. In Fig. 2a the low-frequency mode superposed on the dominant instability easily can be seen where the pressure amplitude and oscillations are

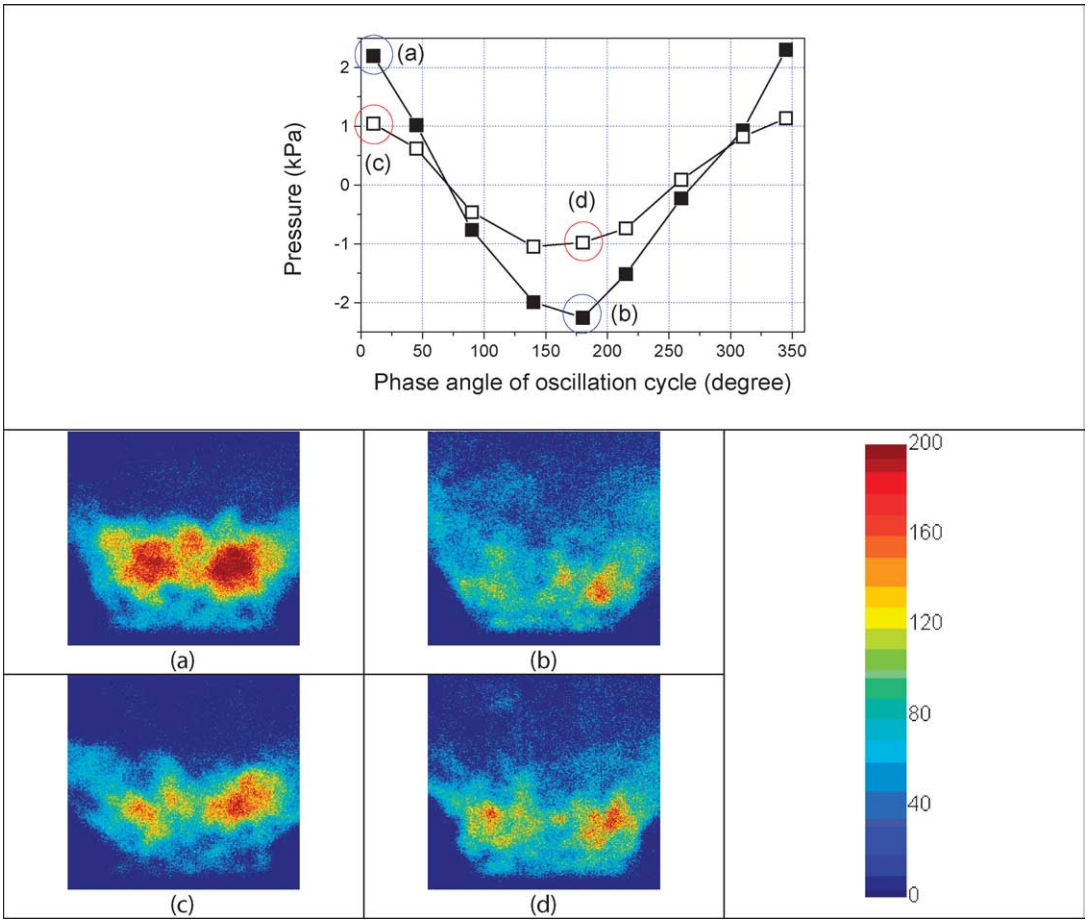
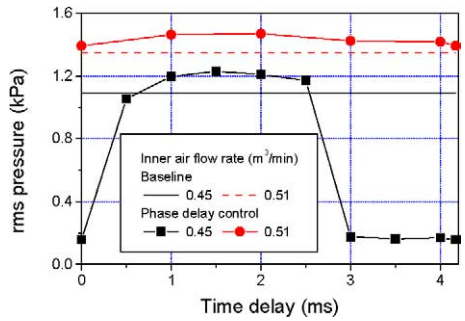


Fig. 9. Influence of low-frequency on the distribution of the CH radical intensity. Inner and annular air flow rates, 0.51 and 1.25 m^3/min, respectively. Primary and secondary fuel flow rate, 3.4 g/s. Overall equivalence ratio, 0.89. Exposure time, 0.1 ms. Solid and open symbols, pressure points at high and low amplitudes during instability cycle.

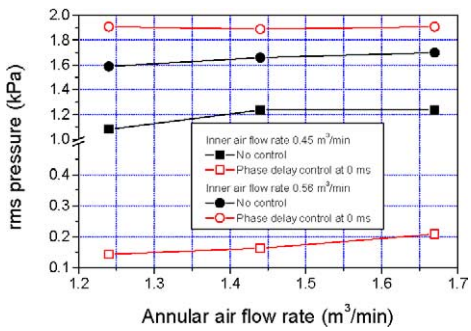
the strongest. One cycle of the pressure oscillation during this time range is shown in Fig. 9 (top) with the solid symbols representing time instances during which phase-locked CH images (averages of 50 images) were taken. Figs. 9a and 9b show the CH images at phase angles of 10° (marked (a) in the pressure cycle shown at top) and 175° (marked (b) in the pressure cycle). It is clearly evident that there is a strong correlation between the pressure and the CH variations; the high-pressure amplitude at 10° phase angle is associated with high CH levels and a more compact flame, while at 175° phase angle, both the pressure and the CH levels are low and the flame is more distributed. The second time interval where phase-locked images are shown (Figs. 9c and 9d) corresponds to time instances where the low-frequency pressure component is a minimum and the pressure oscillation amplitudes are the lowest. One instability cycle, marked by open symbols, is shown in Fig. 9 (top). Again, higher pressures (at 10° phase angle) have higher CH levels and a more compact flame region (Fig. 9c). One should note that the near-minimum pressure levels, observed at 175° phase angle, are lower during the high-pressure oscillation amplitude time periods (marked (b) in Fig. 9, top) than the corresponding value during the low-pressure oscillation period (marked (d) in Fig. 9, top), and therefore the CH levels in Fig. 9d are somewhat higher than those in Fig. 9b. The peak CH levels occur close to the dump plane (bottom line of the image), which corresponds to a pressure antinode.

3.3. *Control of the combustion oscillations by fuel modulations*

In this section, feedback control studies using the traditional approach of time-delay fuel modulation are presented. The centrally located fuel nozzle is used for control purposes. Fuel modulation is explored using both a traditional "on–off" automotive fuel injector valve and a proportional-drive Moog



(a)



(b)

Fig. 10. Phase-delay control with fuel modulation. (a) Inner airflow rates of 0.45 m^3/min (black square) and 0.51 m^3/min (red circle) at same annular airflow rate of 1.25 m^3/min and overall equivalence ratios of 0.92 and 0.89, respectively. (b) Phase-delay control at 0 ms with different inner and annular airflow rates.

valve that permits control of modulation frequency and amplitude. Both valves have frequency responses that exceed the instability frequency range of interest in this study. Typically 10% of the total fuel injected is used for control purposes. These control studies show that under certain conditions such a traditional approach is ineffective.

3.3.1. Liquid fuel modulation with automotive fuel injector

Fig. 10 shows the results of phase-delay control, and at an inner airflow rate of 0.45 m^3/min effective control of the oscillations are achieved (square symbols) for specific time-delay ranges. This behavior is consistent with that reported in the literature. However, when the inner airflow rate is increased to 0.51 m^3/min, no control can be achieved (circular symbols) over the entire range of possible time-delay values. Instead, amplification with nearly a 25% increase in the amplitude over the baseline is observed. This inability to achieve effective control at higher values of the inner airflow rates is due to the control fuel not having enough control authority. Insufficient authority can be related to weak modulation amplitude of the fuel or to the control fuel not reaching the regions of high Rayleigh Index. As shown in Figs. 8d–8f, higher values of the inner airflow rate result in main fuel droplets that are closer to the dump plane and radially displaced outward. Under these conditions, the magnitude of the heat release is considerably higher (as evidenced in Fig. 7d), requiring higher control authority for effective control. It is also likely that the centrally injected control fuel is unable to reach the regions with the strongest heat release dynamics. Both of these factors may play roles in the failure of feedback control with fuel modulations.

To better understand the reasons that make control ineffective at higher inner airflow rates, Fig. 11 is plotted to show the influence of inner airflow rate on the pressure amplitude and heat release intensity. The pressure is measured 5 cm downstream of the combustor dump plane, and the CH chemiluminescence is collected from the region close to the dump plane. Both the rms of the total pressure oscillation and the rms of the bulk mode pressure oscillation (low-frequency modulation) are plotted (Figs. 11a and 11b) in addition to the corresponding CH chemiluminescence values (Figs. 11d and 11e). It is clear that there is a significant increase in the heat release (a factor of 2 or greater in the total heat release fluctuations, and a factor of about 1.4 in the low-frequency heat release fluctuations) as the inner airflow rate is increased from 0.45 to about 0.49 m^3/min, beyond which the CH levels appear to plateau. The low-frequency pressure oscillations also show a sharp increase over this range of inner airflow rates, while the total pressure rms shows a more uniform increase over the entire range of inner airflow rates considered. Fuel-modulation control is essentially ineffective beyond an inner airflow rate of about 0.49 m^3/min, and this coincides with the range where CH fluctuations are very high. Thus, fuel modulation essentially loses control authority in the presence of strong heat release dynamics. It is interesting to note that the trends in the bulk mode oscillations mimic those of the total oscillations, indicating that the oscillations in the combustor are the source that drives the bulk mode in the air delivery chamber. The spectra of the pressure and heat release (Figs. 11c and 11f) show that the oscillation frequencies are unaffected by the inner airflow rates, while the amplitudes reproduce the behavior shown in Figs. 11a and 11d.

A second source for the loss of control authority at the higher inner airflow rate is illustrated in the Mie-scattering images of Fig. 12 where it is shown that the penetration of the control fuel at the higher inner airflow rates is considerably reduced. Thus, the control fuel potentially does not reach the regions of strong heat release dynamics and therefore does not have the requisite control authority.

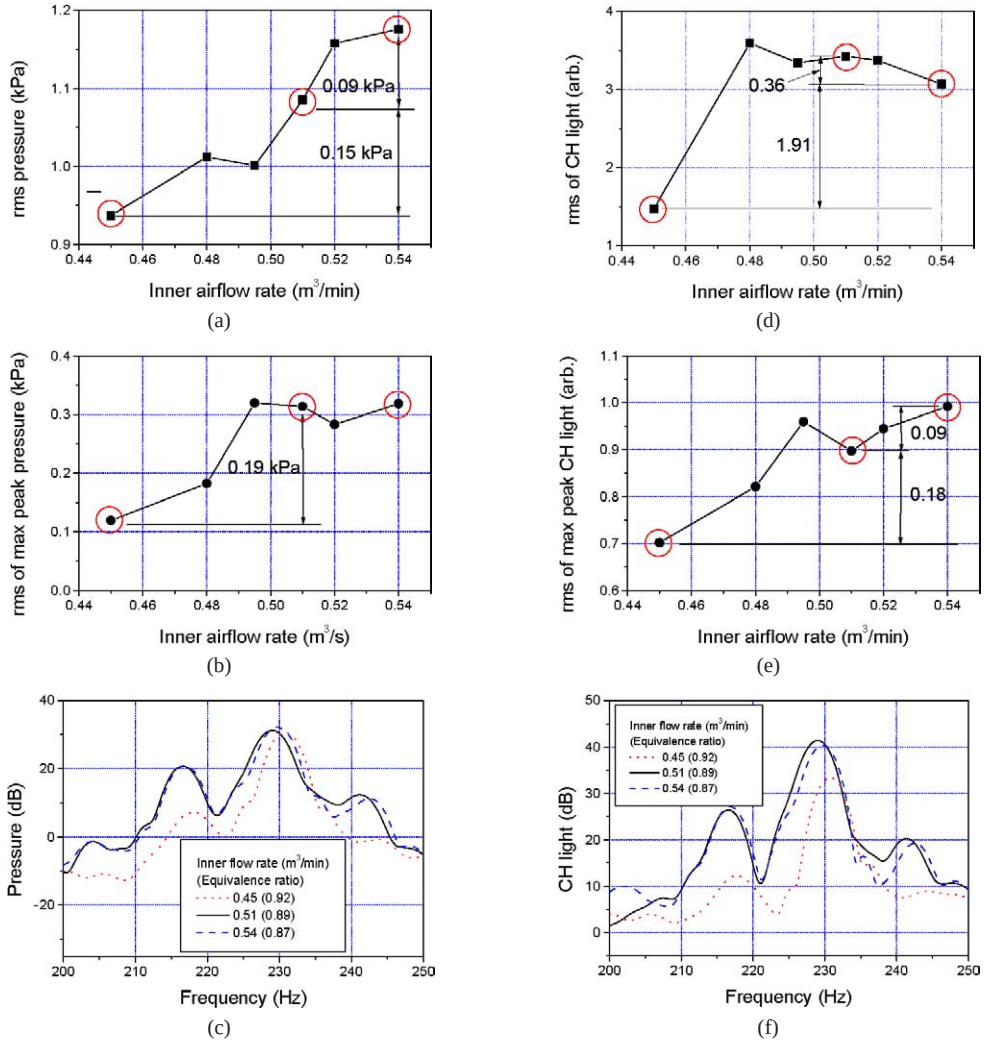
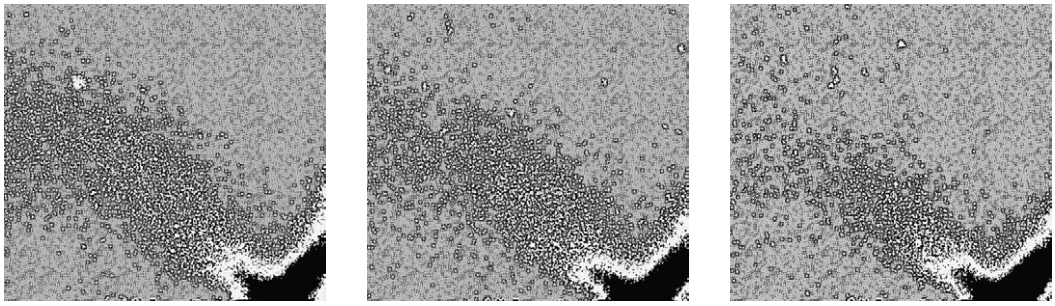
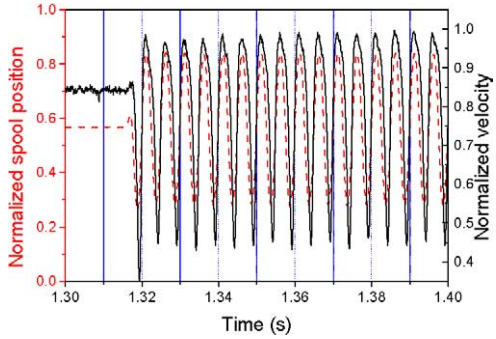


Fig. 11. Variation of amplitude with different inner airflow rates. Annular airflow rate, 1.25 m^3/min. Primary and secondary fuel flow rate, 3.4 g/s. (a) and (d) rms pressure and heat release fluctuations with the variation of inner flow rate. (b) and (e) rms maximum peak pressure and heat release fluctuations. (c) and (f) Pressure and heat release spectra.

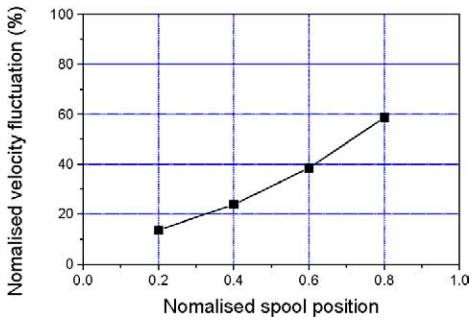


(a) (b) (c)

Fig. 12. Fuel spray visualization. (a)–(c) Control fuel spray. Same annular flow rate, 1.25 m^3/min. Inner air supply of 0.45, 0.51, and 0.54 m^3/min, respectively.



(a)



(b)

Fig. 13. Characteristics of Moog proportional-drive valve at forcing frequency of 200 Hz. (a) Velocity and valve spool position signals. (b) Modulation rate with valve spool gain.

3.3.2. *Liquid fuel modulation with proportional-drive fuel injector*

To overcome the limitations of the automotive fuel injector, a proportional-drive valve manufactured by Moog Inc. was utilized. The proportional-drive valve embodies both gain and phase information and is expected to provide improved control authority through stronger modulation amplitudes.

The proportional-drive valve was first characterized with hot-wire velocity measurements for various preset spool positions that define the gain or amplitude of modulation. These measurements were done at a driving frequency of 200 Hz, which is close to the dominant instability frequency of interest here. The valve gain was adjusted to 60% of the maximum gain and, as shown in Fig. 13a, this gives rise to high-amplitude air modulation rate, with the velocity correlating well with the spool position. Fig. 13b shows the rms of the velocity fluctuation as a function of the spool position, and it is seen that a proportional increase in the gain is achieved with increasing spool position.

Fig. 14 shows that only a modest reduction in the amplitude of the instability was achieved, and the control effectiveness was still poor. There was a maximum of 20% reduction in the amplitude of the pressure oscillations. The poor effectiveness is again

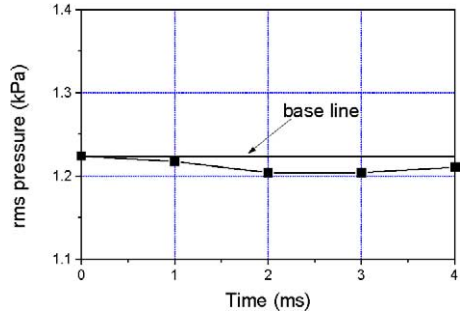


Fig. 14. Phase-delay control by liquid fuel modulation using proportional-drive valve. Inner and annular airflow rates, 0.51 and 1.25 m^3/min, respectively. Primary and secondary fuel flow rate, 3.4 g/s. Overall equivalence ratio, 0.89.

linked to the inability to drive the control fuel to the regions with large Rayleigh Index where the instability originates and to poor control authority.

3.3.3. *Gaseous fuel modulation with proportional-drive fuel injector*

To improve control effectiveness, it was decided to explore the use of propane gas (instead of liquid fuel) as the control fuel. The motivation for using propane gas was based on both the fact that time delays associated with liquid fuel vaporization would not complicate the control process and the empirical observation that the modulation amplitude with the proportional-drive injector was stronger for gas than for liquid. The propane gas was injected from the central nozzle, and different nozzle sizes (1.5, 3, and 6 mm) were used. Fig. 15a shows that feedback control was again relatively ineffective with only marginal reductions in the amplitude of pressure oscillations. Measurements of the CH chemiluminescence at different forcing frequencies at 100 and 300 Hz showed relatively complex CH light spectra, with broadband frequency close to instability and several peaks in the higher frequency ranges (Figs. 15b and 15c). The injected gases are presumably dispersed close to the injection plane, changing the flame structure and anchoring points.

3.4. *Control of combustion oscillations with a high-momentum air-jet*

Based on the results shown in Figs. 10, 14, and 15, it was concluded that a key reason for not achieving control was linked to the inability to deliver the control fuel to the region where the heat release dynamics was generating the instability. The Mie-scattering images in Fig. 12 clearly indicated that the momentum of the inner air-jet played a key role in the dispersion of the injected control fuel. It was also observed that, by controlling the inner airflow rate, the heat release

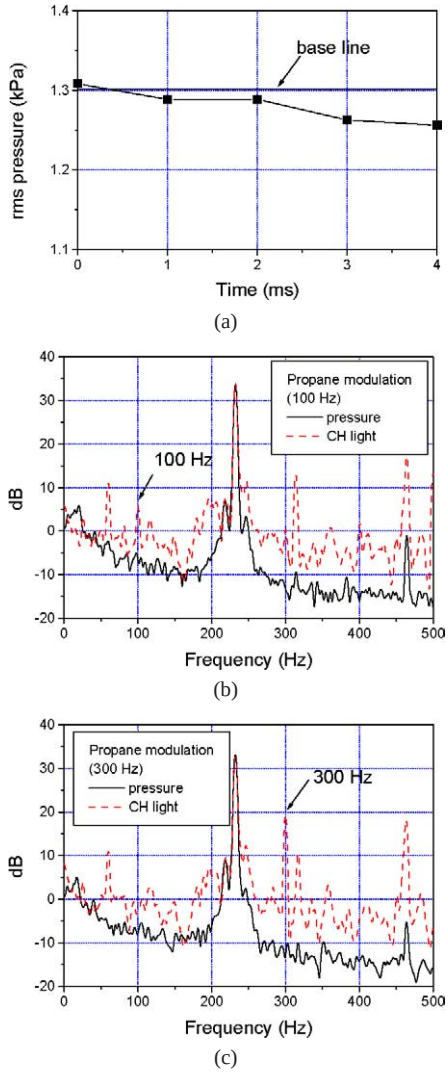


Fig. 15. (a) Phase-delay control by propane gas modulation using proportional-drive valve. (b) Open-loop propane gas modulation at 100 Hz. (c) Open-loop propane gas modulation at 300 Hz. Inner and annular airflow rates, 0.51 and 1.25 m^3/min, respectively. Primary and secondary fuel flow rate, 3.4 g/s. Overall equivalence ratio, 0.89.

could be controlled (Fig. 7) to reduce the amplitude of the pressure oscillations (Fig. 4b). Based on these observations, it was hypothesized that, to achieve effective control for conditions with strong dynamics and/or high inner airflow rates, sufficiently high momentum of the control feed is necessary to penetrate to the appropriate regions and that the control feed could be either fuel or air so long as it is able to disrupt the heat release dynamics driving the instability. From a practical perspective, in a real combustor, it is much more feasible to utilize a high-momentum air-jet for control purposes than a high-momentum fuel-jet, and therefore in the present study a centrally located air-jet was used for control purposes. It should be noted that this approach is different from earlier studies involving acoustic modulation of the entire combustion air stream. With the present approach, a high-momentum control air stream is introduced, and the momentum and location of this jet is tailored to reach the regions in the flow with strong heat release dynamics. As will be shown, only a small amount of air is required to achieve control.

The control air-jet was introduced through the central nozzle used to deliver the control fuel described above. Thus, there is no change in the basic geometry. As examined with propane gas modulation, three nozzle sizes with different airflow rates were explored to determine the appropriate momentum range where control can be achieved. All initial studies were done with a steady air-jet to explore its role in altering the heat release dynamics and thus the combustion instability. This was followed by air-jet modulation studies with the proportional-drive valve.

3.4.1. *Steady air-jet blowing*

The rms of the pressure fluctuations and the mean of the maximum peak pressure are shown in Figs. 16a–16c as a function of the control air-jet flow rate for air-jet nozzle diameters of 1.5, 3, and 6 mm, respectively. With 1.5- and 3-mm nozzle diameters, the pressure oscillations decreased sharply beyond an air-jet flow rate of 12 L/min. However, at these flow rates the velocity exiting the smaller air-jet nozzle diameters is quite high, and any further increases in the air-jet velocity leads to high strain rates, flame lift-off, and local flame quenching. Thus, despite the significant reduction in the pressure oscillation with the air-jet, the window of stable combustion was rather limited with the 1.5- and 3-mm nozzles. With the 6-mm nozzles (Fig. 16c), the reduction in the pressure oscillations is not as dramatic (about 25%), but no flame lift-off or flame quenching is observed due to the lower jet velocities for the same mass flow rates. Fig. 16c also shows the corresponding rms and peak values at the same overall equivalence ratio but without any air-jet (open symbols). Thus, a considerable reduction in pressure oscillations is achieved with only a small amount of air (15 L/min, less than 1% of total air supply for main combusting air) and is required to achieve control. With the addition of this small amount of air, the change in the overall equivalence ratio, for the flow conditions corresponding to Fig. 16, is only 0.007, and therefore the control achieved is not a consequence of the operating conditions (overall equivalence ratio) shifting beyond the instability range. The range of the control air supply flow rates needed to favorably influence the combustion dynamics is determined experimentally by

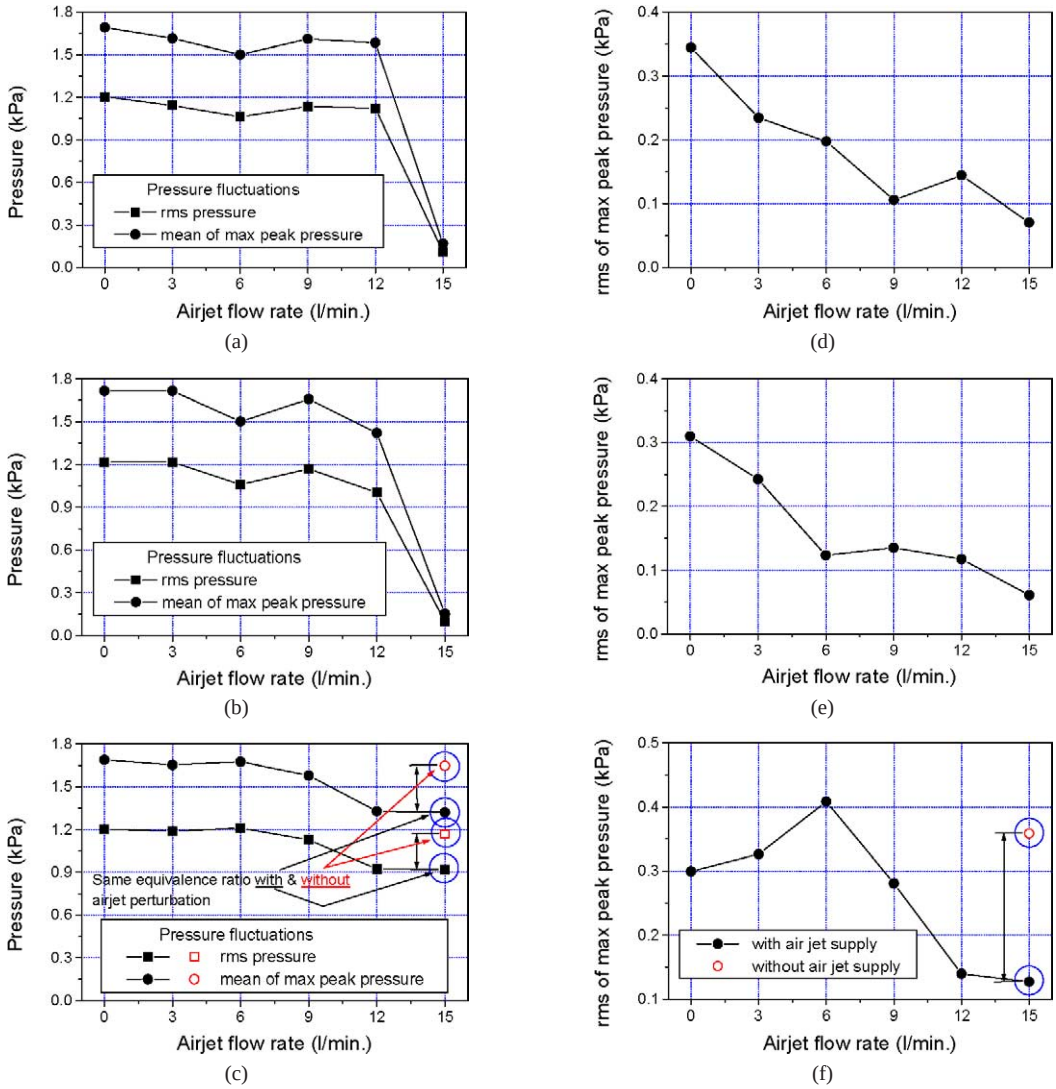


Fig. 16. Influence of nozzle diameter for different air-jet flow on the combustion oscillations. Inner and annular airflow rates, 0.51 and 1.25 m^3/min, respectively. Primary and secondary fuel flow rate, 3.4 g/s. (a)–(c) rms pressure fluctuations and means of maximum peak pressure with nozzle diameters of 1.5, 3, and 6 mm, respectively. (d)–(f) rms maximum peak pressures with nozzle diameters of 1.5, 3, and 6 mm, respectively. Overall equivalence ratios with air-jet flow rates of 0, 3, 6, 9, 12, and 15 L/min: 0.886, 0.885, 0.883, 0.882, 0.880, and 0.879, respectively.

varying the control air-jet supply rates (Fig. 16). Arguably, lower control flow rates would be needed with modulation.

Figs. 16d–16f show the rms of the low-frequency pressure oscillation associated with the bulk mode of the air delivery nozzle. The higher-frequency dominant instability mode is filtered out of the pressure signal before computing the low-frequency pressure rms. It is seen that the steady air-jet significantly reduces (by a factor of two to five) the low-frequency oscillation, and this reduction is a major factor in the reductions observed in the total rms pressure oscillations.

Mie-scattering images of the fuel droplets for different air-jet nozzle sizes (1.5, 3, and 6 mm) with and without the control air-jet injection are shown in Fig. 17. The control air-jet is located along the centerline, which corresponds to the right-hand boundary of the image. As seen earlier in Fig. 8, the atomized droplets are directed radially outward and upward. With decreasing nozzle sizes and increasing velocity of the control air-jet, there appears to be increased entrainment of the droplets toward the centerline. This can potentially disperse the fuel droplets more uniformly and alter the heat release dynamics in a ben-

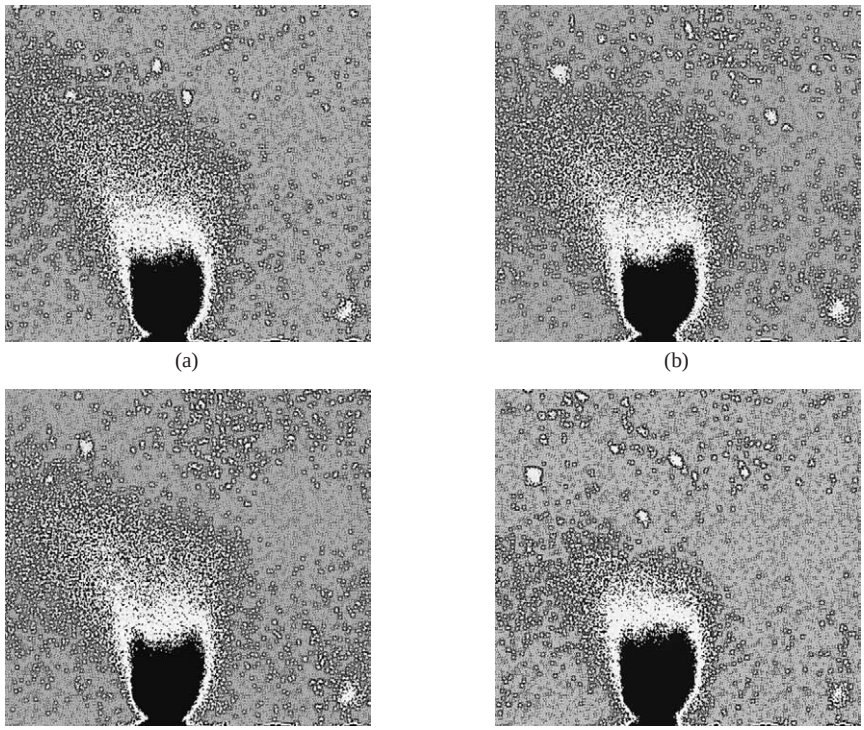


Fig. 17. Fuel spray visualization with air-jet supply from different sizes of nozzles. Inner and annular airflow rates, 0.51 and 1.25 m^3/min, respectively. Air-jet flow rate, 15 L/min. (a) Without air-jet supply. (b)–(d) 1.5, 3, and 6 mm, respectively.

eficial manner. However, as noted earlier, the high velocities of the control air-jet influence the flame anchoring adversely, and flame lift-off occurs beyond 15 L/min of control air. With the largest air-jet nozzle (Fig. 17d), the air-jet momentum is lower but its initial width is greater, relative to the 1.5- and 3-mm-diameter nozzles. Therefore, the control air-jet, instead of entraining droplets inward, penetrates conically outward into the region occupied by the primary fuel droplets. This again alters the flame and heat release dynamics to reduce oscillations; furthermore, there is no adverse impact on flame lift-off. There-

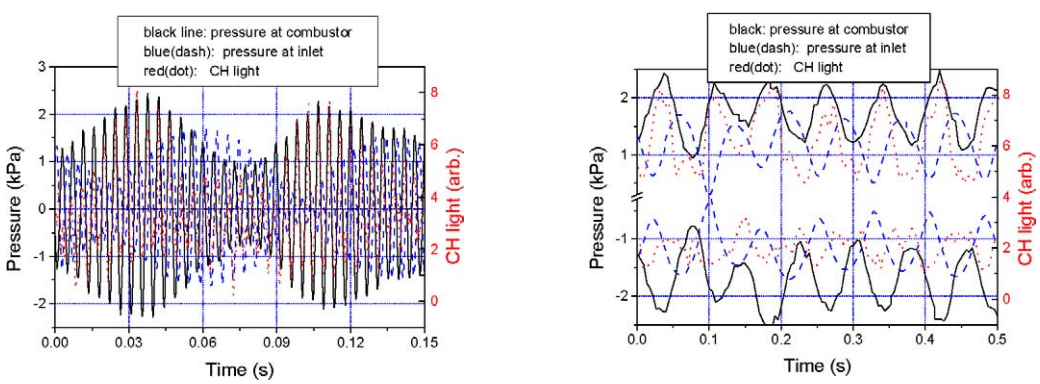


Fig. 18. Influence of air-jet on the combustion oscillations of low- and acoustic-frequency modes. Inner and annular airflow rates, 0.51 and 1.25 m^3/min, respectively (overall equivalence ratio, 0.886). Primary and secondary fuel flow rate, 3.4 g/s. Air-jet flow rate, 15 L/min (overall equivalence ratio, 0.879). (a)–(d) and (e)–(h) Without and with air-jet supply, respectively. (a) and (e) Signals of pressure at the combustor and the upstream end of annular air stream duct and of CH light at the combustor. (b) and (f) Maximum and minimum peaks of pressure and CH light. (c) and (g) Cycle variation of the instability. (d) and (e) Time variation of the instability between the combustor and the upstream end of annular air stream duct.

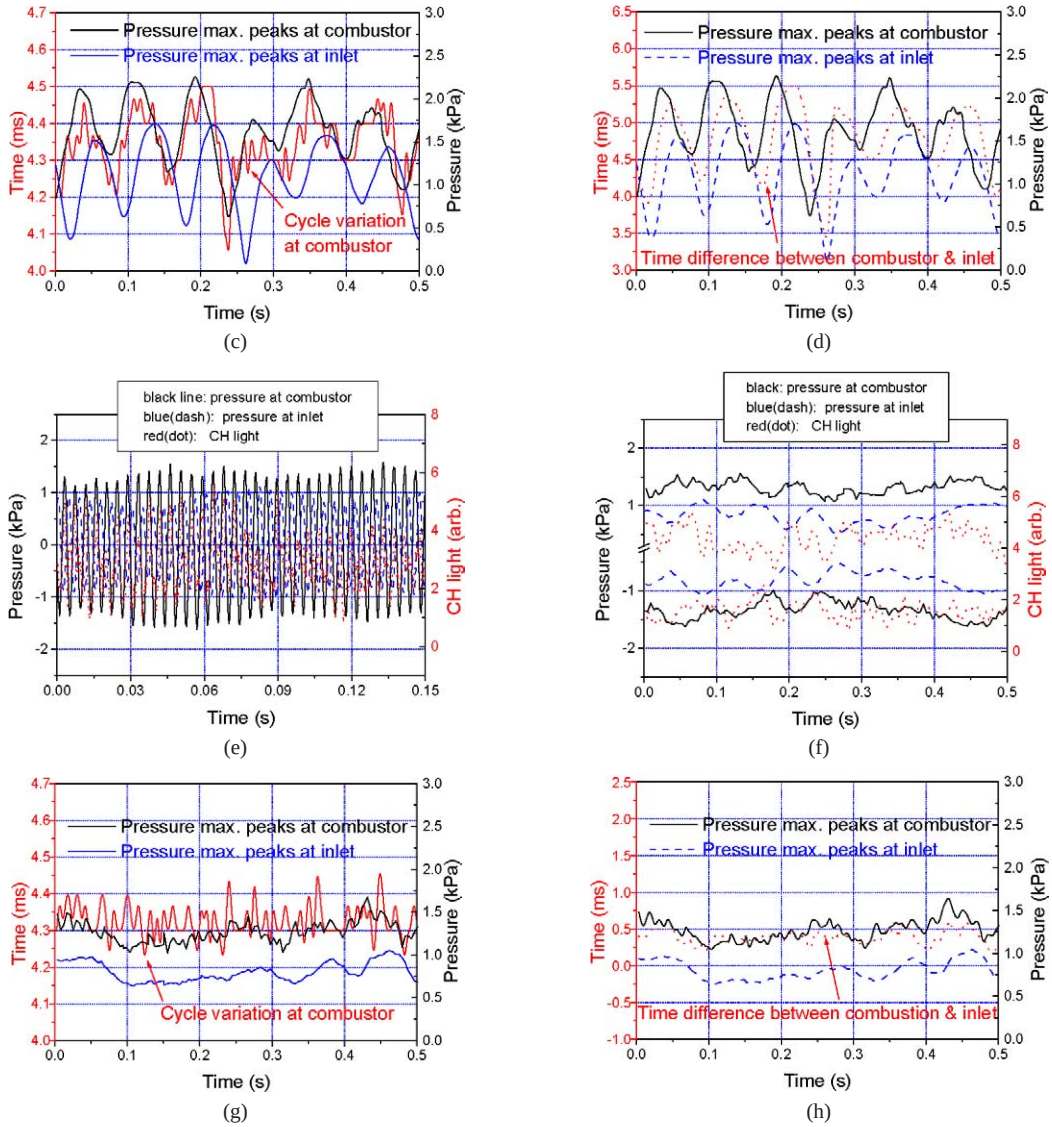


Fig. 18. (Continued.)

fore the nozzle size has to be tailored to achieve the optimal momentum ratio, which in turn will favorably alter the droplet/heat release distributions to decrease the combustion dynamics.

Figs. 18a–18d show the pressure signal in the combustor and in the air delivery chamber without control air-jet injection. Also shown in the figures are the CH intensity variations with time. The corresponding plots with steady control air-jet injection are shown in Figs. 18e–18h. The air-jet interferes with the combustion oscillations and leads to significant reductions from 5 to 3 kPa in the maximum peak-to-peak amplitude of the pressure in the combustor (compare Figs. 18a and 18b with Figs. 18e and 18f) and a corresponding reduction in CH emission from 9 to 4 (arbitrary units). Figs. 18c and 18g plot the low-frequency variation of the pressure peaks and the cycle-to-cycle variation in the time period of the dominant instability. It can be seen that the cycle-to-cycle variation in the instability cycle time period is nearly 0.5 ms (associated with 42° of the period of the instability in Fig. 18c), which provides a measure of the phase jitter or spectral broadening in the dominant instability. This phase jitter is driven by the low-frequency flame movement induced by the bulk mode of the air delivery chamber. Fig. 18g shows that with the introduction of the air-jet, both the low-frequency mode and the cycle-to-cycle variation in the instability cycle time period are considerably reduced. Thus, with the steady control air-jet, both the pressure os-

cillations and the spectral broadening are reduced. Figs. 18d and 18h show the low-frequency variations in the pressure peak in the combustor and in the upstream air delivery chamber and the time difference between these signals. The time difference between the signals is a measure of the suitability of using a transducer located in the air delivery chamber for control purposes. A maximum 5.25 ms delay between the two pressure signals is observed, reflecting a time delay that exceeds a full period of the dominant instability. Thus the existence of the low-frequency mode introduces significant time delays between measurements made in the combustor and those made upstream in the air delivery chamber. With control air injection, since the low-frequency mode is considerably reduced, the time delay between the two pressure signals is reduced by an order of magnitude to 0.5 ms.

As shown earlier in Fig. 3, a half-wave longitudinal mode is established in the air delivery chamber, which combined with the quarter-wave in the combustor results in a three-quarter-wave acoustic mode in the system. The half-wave mode in the air delivery chamber is shown in Figs. 19a and 19b. Air-jet blowing leads to reductions of the rms pressure fluctuations of 30 and 66% in the dominant instability and low-frequency modes, respectively. Fig. 19c shows the phase relationship of the dominant frequency between combustor and various locations along the air delivery chamber. These results are consistent with those shown in Figs. 18d and 18h and show a factor of four reduction in the time delay between the two signals.

Fig. 20 shows the pressure and CH emission values without and with air-jet injection. Also shown are phase-locked images at different time instances identified as Figs. 20a–20e in the time traces. Figs. 20a–20c correspond to the baseline case with no air-jet, while Figs. 20d and 20e represent cases with air-jet injection. The time-averaged images without and with air-jet injection are shown in Figs. 20f and 20g, respectively. With air-jet injection, subtle structural changes in the CH patterns can be observed: (1) the intensity levels (and the associated dynamics) have lower amplitudes and (2) the distribution of CH intensity is more uniform with a narrower spatial extent.

3.4.2. Modulated high-momentum air-jet

The discussions in the previous section identified the potential benefits of using a steady high-momentum air-jet. Significant reductions in the low-frequency oscillations were observed with this approach. In this section, feedback loop control using a modulated high-momentum air-jet is explored. The air-jet is modulated using the proportional-drive Moog valve described earlier. Control studies were done for flow conditions of inner and annular airflow rates of 0.51 and 1.25 m^3/min, respectively, for which, as described earlier, effective control was not achieved (no reduction or less than 10% reduction in the pressure rms values) using fuel-modulation strategies.

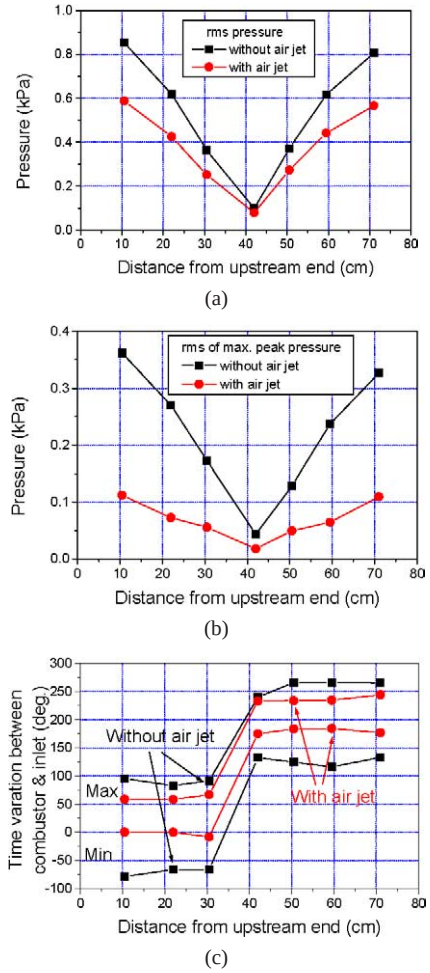


Fig. 19. Waveform of the dominant acoustic mode (230 Hz) and bulk mode (13 Hz) in the annular air stream duct with (red circle) and without (black square) air-jet injection. Inner and annular airflow rates, 0.51 and 1.25 m^3/min, respectively (overall equivalence ratio, 0.886). Primary and secondary fuel flow rate, 3.4 g/s. Air-jet flow rate, 15 L/min (overall equivalence ratio, 0.879). (a) Waveform with dominant mode. (b) Waveform with low-frequency mode. (c) Phase relationship between the combustor and the locations along annular air stream duct.

Fig. 21 shows that the baseline (defined as conditions with steady air-jet blowing; comparisons with no-blowing conditions would be even more favorable) rms pressure levels were reduced to 0.78 kPa, 70% of that observed with fuel modulations. This is a direct consequence of local injection of air with suffi-

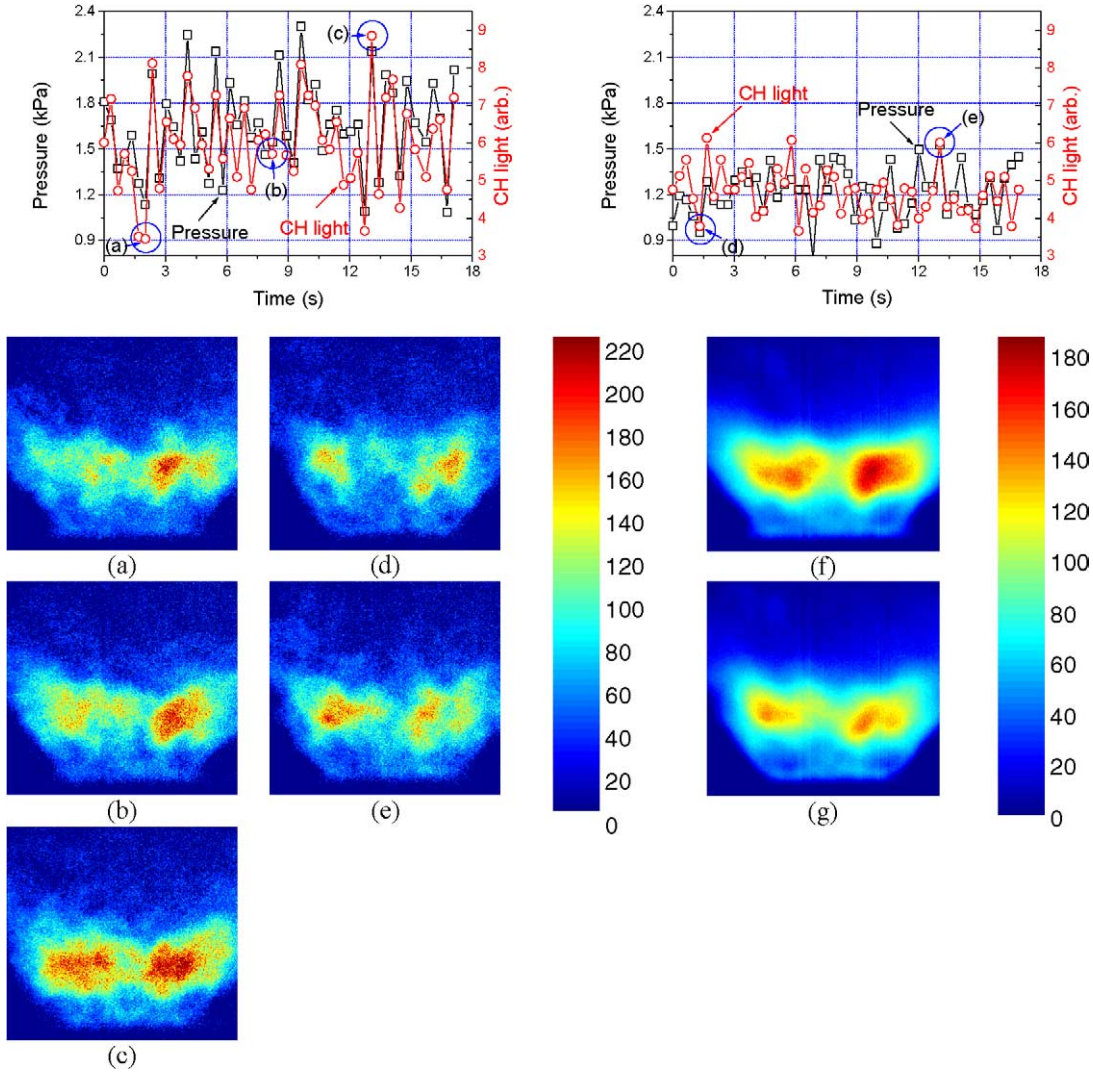


Fig. 20. CH image with and without air-jet supply. Inner and annular airflow rates, 0.51 and 1.25 m^3/min, respectively. Primary and secondary fuel flow rate, 3.4 g/s. Air-jet flow rate, 15 L/min. (Top) Pressure and CH values synchronized on the pressure maximum without (left) and with (right) air-jet supply. (a)–(e) Instantaneous CH images indicated on top graphs. Exposure time, 0.1 ms. (f) and (g) Time-averaged CH images synchronized on the 50 points of top graphs. Exposure time, 4 ms.

cient momentum through the control nozzle to change the reactant mixing process in the regions of positive Rayleigh Index, so that the turbulent flame structure and heat release dynamics are completely changed. With time-delay feedback control, the instability was essentially completely suppressed with pressure rms levels in the range of 0.1 kPa (Fig. 21a). This is nearly a 90% reduction (eightfold) in the instability amplitude as opposed to less than a 10% reduction with the traditional fuel-modulation control. Figs. 21b and 21c show the pressure and CH spectra and the significant reduction in the amplitude of the dominant instability. The CH spectrum with air-jet modulation shows several peaks in the range from 200 to 350 Hz and is a reflection of the changes in the heat release dynamics introduced by the air-jet. However, no significant peak is observed at the dominant instability mode of 230 Hz, which explains the significant reduction in the acoustic instability amplitudes.

4. Concluding remarks

The characteristics of large-amplitude combustion oscillations in a swirl-stabilized spray combustor are investigated through measurements of wall pressure, CH chemiluminescence, and CH imaging. The combustion oscillations are dominated by an acoustic

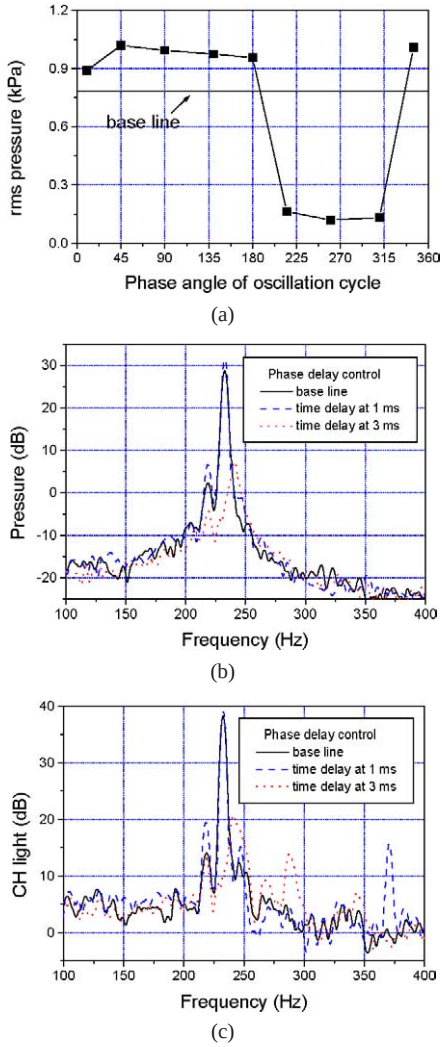


Fig. 21. Phase-delay control by air-jet modulation. Inner and annular airflow rates, 0.51 and 1.25 m^3/min, respectively. Primary and secondary fuel flow rate, 3.4 g/s. Air-jet flow rate, 15 L/min. Overall equivalence ratio, 0.879. (a) Phase-delay control. (b) Pressure spectra with and without control. (c) CH light spectra with and without control.

three-quarter-wave mode in the range of 225–235 Hz depending on flow conditions that occurred in the entire system including the combustor and the air delivery chamber. A low-frequency bulk mode (13 Hz) associated with the upstream air delivery chamber is also established. The low-frequency oscillations are likely to be associated with periodic flame movement. This low-frequency mode leads to a cycle-to-cycle variation or phase jitter in the time period of the dominant instability mode, which makes feedback control of the instability more difficult.

The inner airflow rate is shown to control the heat release dynamics, and when the inner airflow rate is increased beyond 0.49 m^3/min, the heat release dynamics become sufficiently strong to make control authority with fuel modulation ineffective. Under these conditions, no significant reduction in the pressure oscillations can be achieved with either liquid fuel or gaseous fuel modulation. Mie-scattering images of the fuel droplets show that, at the higher values of the inner airflow rate, the control fuel dispersion is limited and does not penetrate to the regions where the heat release dynamics are strong. This is believed to be a reason for poor control authority.

An improved strategy of using a high-momentum control air-jet is shown to be effective in reducing both the bulk mode and the longitudinal mode pressure oscillations. The control air-jet is injected centrally, and suitable selection of the jet diameter and momentum are shown to be important for effective control. Only a small amount of control air-jet (with a corresponding change in equivalence ratio of 0.007) is needed for effective control. The control air-jet is shown to interfere with the droplet and the heat release distributions to beneficially impact the combustion dynamics. Even with steady air-jet blowing, reductions in the pressure oscillations are observed, particularly in the magnitude of the bulk mode component. Air-jet injection is also shown to reduce both the phase jitter in the dominant instability and the phase lag between the pressure signals in the combustor and the air delivery chamber. With a modulated control air-jet, nearly an eightfold (or 90%) reduction in the peak pressure oscillation amplitude is achieved. This is in contrast to a maximum reduction by 10% of the pressure oscillations with fuel modulation control.

Acknowledgments

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References

[1] J.A. Austin, J.R. Tilson, I.R.I. McKenzie, in: Proc. RTO AVT Symposium on Active Control Technology for Enhanced Performance Operation Capabilities of Military Aircraft, Land Vehicles and Sea Vehicles, Braunschweig, 2000.

[2] G.A. Richards, M.C. Janus, ASME J. Eng. Gas Turbines Power 120 (1998) 294–302.

[3] J.O. Keller, L. Vaneveld, D. Korschelt, G.L. Hubbard, A.F. Ghoniem, J.W. Daily, A.K. Oppenheim, AIAA J. 20 (1981) 254–262.
[4] J.M. Cohen, N.M. Rey, Jacobson, C.A. Anderson, T.J. Rosfjord, in: Proc. RTO AVT Symposium on Gas Turbine Engine Combustion, Emissions and Alternative Fuels, Lisbon, 1998, Paper No. 38.
[5] T. Poinsot, D. Veynante, F. Bourienne, S. Candel, E. Esposito, J. Surget, Proc. Combust. Inst. 22 (1988) 1363–1370.
[6] G. Isella, C. Seywert, F.E.C. Culick, E.E. Zukoski, Combust. Sci. Technol. 126 (1997) 381–388.
[7] P.J. Langhorne, J. Fluid Mech. 193 (1988) 417–443.
[8] S.R.N. De Zilwa, S. Sivasegaram, J.H. Whitelaw, Flow Turbulence Combust. 63 (1999) 395–414.
[9] D. Bradley, P.H. Gaskell, X.J. Gu, M. Lawes, M.J. Scott, Combust. Flame 115 (1998) 515–538.
[10] E. Gutmark, T.P. Parr, D.M. Hanson-Parr, K.C. Schadow, AIAA J. 29 (12) (1991) 2155–2162.
[11] S.R.N. De Zilwa, J.H. Uhm, J.H. Whitelaw, Combust. Sci. Technol. 160 (2000) 231–258.
[12] S.R.N. De Zilwa, I. Emiris, J.H. Uhm, J.H. Whitelaw, Proc. R. Soc. London A 457 (2001) 1915–1949.
[13] J.H. Uhm, Nature and Control of Combustion Oscillations in Sudden-Expansion Flows, Ph.D. thesis, Imperial College London, 2001.
[14] F.E.C. Culick, in: AGARD Conference on Combustion Instabilities in Liquid-Fuelled Propulsion System, Neuilly Sur Seine, 1988, Proc. No. 450 (1).
[15] K.C. Schadow, E. Gutmark, Prog. Energy Combust. Sci. 18 (1992) 117–132.
[16] S.M. Candel, Proc. Combust. Inst. 24 (1992) 1277–1296.
[17] K.R. McManus, T. Poinsot, S.M. Candel, Prog. Energy Combust. Sci. 19 (1993) 1–29.
[18] W. Lang, T. Poinsot, S. Candel, Combust. Flame 70 (1987) 281–289.
[19] S. Hoffmann, G. Weber, H. Judith, J. Hermann, A. Orthmann, in: Proc. RTO AVT Symposium on Gas Turbine Engine Combustion, Emissions and Alternative Fuels, 1998, Paper No. 40.
[20] D.U. Dampos-Delgado, K. Zhou, D. Allgood, S. Acharya, Combust. Sci. Technol. 175 (1) (2003) 1–27.
[21] J.H. Uhm, S. Acharya, AIAA-2003-0339, Reno, NV, USA, Jan. 6–9, 2003.
[22] S. Sivasegaram, J.H. Whitelaw, ASME Active Control Noise Vibrat. DSC-38 (1992) 69–74.
[23] S. Sivasegaram, R.-F. Tsai, J.H. Whitelaw, Combust. Sci. Technol. 105 (1995) 67–83.
[24] K.H. Yu, in: Proc. RTO AVT Course on Active Control of Engine Dynamics, Brussels, Belgium, May, 2001.
[25] M. Fleifil, A.M. Annaswamy, J.P. Hathout, A.F. Ghoniem, Combust. Sci. Technol. 133 (1998) 227–260.
[26] K.J. Wilson, E. Gutmark, K.C. Schadow, AIAA Paper 91-0368, Reno, NV, Jan. 7–10, 1991.
[27] R. Bhidayasiri, S. Sivasegaram, J.H. Whitelaw, Thermofluids Section Report TF/98/08, Imperial College of Science Technology and Medicine, London, 1998.